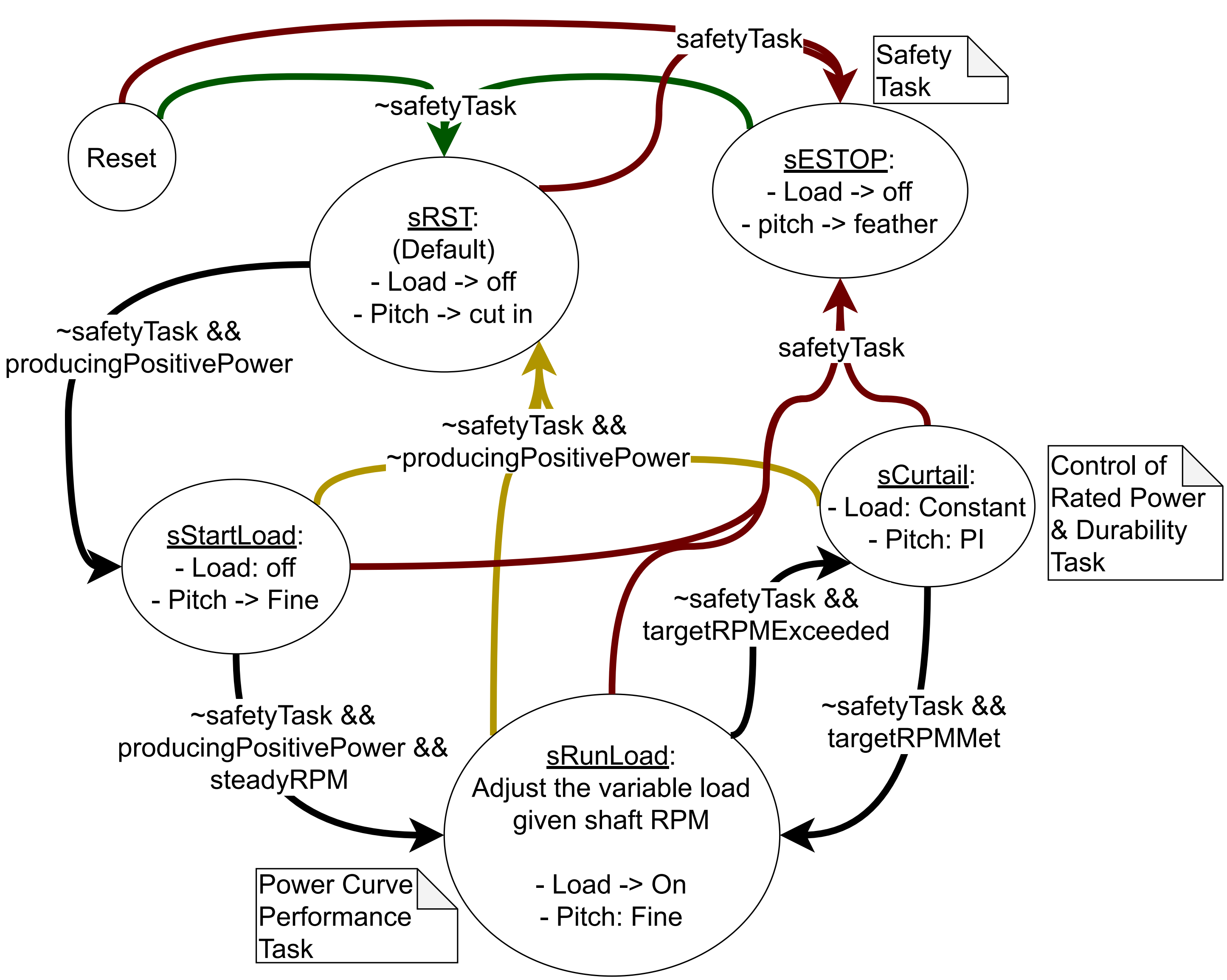
2026 FSM Draft

Implementation Notes
- Although all outputs have been depicted as per state (More), they will moved to state transitions (Mealy) in the implementation
- This may be split into to parallel software FSMs, one per controller, which may improve reliability.
- Last year, we ran the FSM at 9 Hz
- This loop may be overridden manually for testing
- Backfeed controls have been moved to hardware as of last year

Notes for Testing:
- May need to limit how fast the variable load can change to promote stabiltiy
- May need to watch for undervoltage condition, but this may be unneeded and add complexity



Nacelle Inputs & Variables
steadyRPM = (dRPM/dt < 20 RPM/s)
targetRPMExceeded = (RPM > targetRPM)
targetRPMMet = RPM < (targetRPM - sHysteresisWindow)
Note: Hysteresis may not be ideal and should not be needed if PID is well tuned

Load Inputs & Variables
safetyTask = (ESTOP || loadDisconnect)
loadDisconnect = PCCCurrent < (0 + iHysteresis)
producingPositivePower = PCCVoltage > (backFeedVoltage + vHysteresisWindow/2)
Note: Previous versions included a fixed stage references of > 6 V (?).